

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

First/Second Semester B.E. (E&E)(Credit System) Supplementary Examinations

Supplementary Examinations - December 2025

21EE104/20EE104/17EE105 – BASIC ELECTRICAL ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** and **Unit – II** each and **one full** question from **Unit – III**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Explain the terms i)Average value, ii)RMS value, iii)Form factor, and iv)Peak factor, of sinusoidally varying voltage and current. | 6 | L*2 | 1 | 1,2 |
| b) Derive an expression for calculating line and phase quantities of Star connected system | 7 | L2 | 2 | 1,2 |
| c) An AC Circuit consist of a pure resistance of 10 ohms and is connected across an AC supply of 230 V, 50Hz. Calculate current, power consumed, equation for voltage and current. | 7 | L3 | 2 | 1,2 |
| 2. a) Using mesh analysis, find the voltage drop across 3Ω resistor in the figure 2a. | | | | |

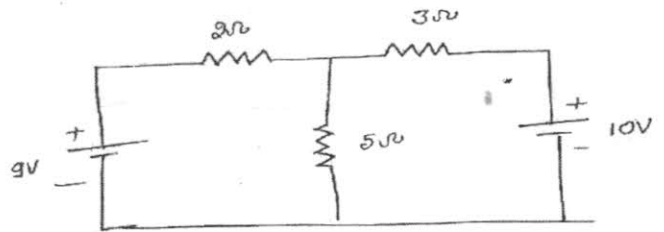


Figure 2a.

- | | | | | |
|--|---|----|---|-----|
| | 7 | L3 | 1 | 1,2 |
| b) Explain the following basic concepts of an alternating quantity,
i) Waveform
ii) Cycle
iii)Time Period (T)
iv) Frequency (f)
v) Instantaneous Value
vi) Amplitude
vii) Phase | 7 | L2 | 1 | 1,2 |
| c) Derive an equation which relates line and phase quantities of Delta-connected system. | 6 | L3 | 2 | 1,2 |
| 3. a) With neat circuit diagram, and waveforms derive the relationship between voltage and current in a RC circuit. Also derive the expression for power consumed. | 6 | L3 | 1 | 1,2 |
| b) Using nodal analysis, find the voltages at node b for the circuit shown in figure 3b. | | | | |

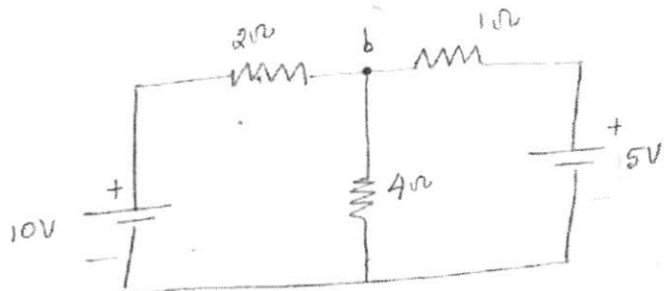


Figure 3b.

P.T.O.

7 L3 1 1,2

- c) Derive an expression for RMS value of current from the basic equations.

7 L2 1 1,2

Unit – II

4. a) Derive the EMF equation of single-phase Transformer.
 b) Explain the principle of operation of Synchronous Motors.
 c) Derive an expression to find the generated RMS value of emf/phase for three phase synchronous generator.
5. a) A 250 kVA, 11000 V/400 V, 50 Hz single-phase transformer has 80 turns on the secondary. Calculate: (i) the primary and secondary current; (ii) the approximate number of primary turns; (iii) the maximum value of the flux.
 b) Explain different losses occurring in a transformer when it is excited by an AC source.
 c) With neat diagram explain the construction of DC machine.
6. a) The primary and secondary windings of a 500 kVA transformer have resistances of 0.42Ω and 0.0019Ω respectively. The primary and secondary voltages are 11000 V and 400 V respectively and the core loss is 2.9 kW, assuming the power factor of the load to be 0.8. Calculate the efficiency on (a) full load; (b) half load.
 b) Derive the condition for maximum efficiency of a single-phase transformer.
 c) From the fundamentals, derive the EMF equation of a DC generator.

6 L3 3 1,2

7 L2 4 1,2

7 L3 4 1,2

7 L3 3 1,2

6 L3 3 1,2

7 L2 4 1,2

7 L3 3 1,2

6 L2 3 1,2

7 L3 4 1,2

Unit – III

7. a) What is meant by earthing? With neat diagram explain plate earthing.
 b) Explain the concept of rotating magnetic field in three phase induction motor.
 c) Explain the importance of Fuse and MCB in domestic wiring.
8. a) With neat diagram explain concealed conduit wiring.
 b) Explain the necessity of Starter for 3 Phase Induction Motor.
 c) With neat schematic diagram and vector diagram, explain working of single phase induction motor.

7 L2 5 1,2

7 L2 5 1,2

6 L2 5 1,2

7 L2 5 1,2

6 L2 5 1,2

7 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
